

LogicConnect: Evidence Atlas

Detailed supporting results and plots for internal review and presentation. This artifact is separate from the DATE submission. LC denotes the unified construction; LC-D is a distinct dense specialization. All local values come from the frozen evidence snapshot. Reported literature values retain separate labels and source records. No new training or inference was performed.

Dense held-out accuracy: all methods and seeds

T throughout; three seeds. LogicConnect-D is the distinct dense specialization. A best-of-variants test maximum is not an evaluated method-selection policy.

Model	Gates	Method	Mean $\pm$ SD (%)	Seeds 0, 1, 2 (%)	Source
MNIST	8000	Random	91.273 $\pm$ 0.217	91.190, 91.520, 91.110	REP
MNIST	8000	LogicConnect	91.937 $\pm$ 0.137	91.790, 91.960, 92.060	OUR
MNIST	8000	LogicConnect-D	91.907 $\pm$ 0.307	91.570, 91.980, 92.170	OUR
Fashion-MNIST	16000	Random	85.197 $\pm$ 0.261	85.110, 85.490, 84.990	REP
Fashion-MNIST	16000	LogicConnect	85.717 $\pm$ 0.453	85.340, 85.590, 86.220	OUR
Fashion-MNIST	16000	LogicConnect-D	85.913 $\pm$ 0.356	86.280, 85.890, 85.570	OUR
CIFAR-10 S	48000	Random	48.913 $\pm$ 0.346	48.840, 49.290, 48.610	REP
CIFAR-10 S	48000	LogicConnect	52.097 $\pm$ 0.630	52.070, 51.480, 52.740	OUR
CIFAR-10 S	48000	LogicConnect-D	52.390 $\pm$ 0.295	52.400, 52.680, 52.090	OUR
CIFAR-10 M	512000	Random	54.097 $\pm$ 0.171	53.900, 54.210, 54.180	REP

Dense held-out accuracy: all methods and seeds (continued)

Model	Gates	Method	Mean $\pm$ SD (%)	Seeds 0, 1, 2 (%)	Source
CIFAR-10 M	512000	LogicConnect	58.653 $\pm$ 0.168	58.800, 58.690, 58.470	OUR
CIFAR-10 M	512000	LogicConnect-D	58.357 $\pm$ 0.253	58.450, 58.070, 58.550	OUR
CIFAR-10 L	1280000	Random	55.870 $\pm$ 0.308	55.820, 55.590, 56.200	REP
CIFAR-10 L	1280000	LogicConnect	60.463 $\pm$ 0.348	60.080, 60.550, 60.760	OUR
CIFAR-10 L	1280000	LogicConnect-D	61.073 $\pm$ 0.463	60.540, 61.310, 61.370	OUR

Dense paired effects against native random

T, paired over the complete three-seed set. Intervals are not adjusted for multiple comparisons. Both variants are retained.

Model	Method	Gain (pp)	Paired 95% CI	Wins / 3	Raw gains (pp)
MNIST	LogicConnect	0.663	[0.015, 1.311]	3	0.600, 0.440, 0.950
MNIST	LogicConnect-D	0.633	[-0.290, 1.557]	3	0.380, 0.460, 1.060
Fashion-MNIST	LogicConnect	0.520	[-1.016, 2.056]	3	0.230, 0.100, 1.230
Fashion-MNIST	LogicConnect-D	0.717	[-0.284, 1.717]	3	1.170, 0.400, 0.580
CIFAR-10 S	LogicConnect	3.183	[0.772, 5.595]	3	3.230, 2.190, 4.130
CIFAR-10 S	LogicConnect-D	3.477	[3.265, 3.688]	3	3.560, 3.390, 3.480
CIFAR-10 M	LogicConnect	4.557	[3.781, 5.332]	3	4.900, 4.480, 4.290
CIFAR-10 M	LogicConnect-D	4.260	[3.371, 5.149]	3	4.550, 3.860, 4.370
CIFAR-10 L	LogicConnect	4.593	[3.721, 5.466]	3	4.260, 4.960, 4.560
CIFAR-10 L	LogicConnect-D	5.203	[3.959, 6.447]	3	4.720, 5.720, 5.170

Convolutional accuracy and training cost

S/M are nine-channel, paper-faithful architectures. V and T are separate observations. Recorded wall times include evaluation and depend on execution conditions. GPU is peak PyTorch allocation.

Size	Method	Scope	n	Accuracy (%)	Time (h)	GPU (GiB)	Gate functions
S	Random	T	1	57.370	4.975	1.831	83552
S	LogicConnect	T	1	60.630	4.951	1.831	83552
M	Random	T	1	69.570	25.470	14.614	668416
M	LogicConnect	T	1	71.650	35.686	14.614	668416
S	Random	V	3	59.820 ± 1.130	4.966	1.831	83552
S	LogicConnect	V	3	61.053 ± 1.041	4.970	1.831	83552

LILogic architectures: achieved and reported values

LILogic v2 (2026), seven thresholds. Published time is contextual because execution platforms and implementations differ. Reported peak GPU allocation is unavailable. Top-32 local n=1; fixed methods n=3.

Size	Method	n	Local T (%)	Local min	Local GiB	Params (M)	Reported T (%)	Reported min
M	Random	3	49.010 ± 0.426	14.622	0.474	1.024	49.17 ± 0.14	2.9
M	LogicConnect	3	52.543 ± 0.296	14.593	0.474	1.024	–	–
M	Top-32	1	57.840	37.808	8.100	5.120	57.28 ± 0.30	15.3
L	Random	3	55.333 ± 0.469	15.202	1.557	4.096	54.76 ± 0.27	16.3
L	LogicConnect	3	60.193 ± 0.286	18.472	1.557	4.096	–	–
L	Top-32	1	62.030	350.635	24.861	20.480	60.98 ± 0.19	297.2

Derived routing trade-offs

Ratios use locally observed means. These quantities describe an accuracy/resource trade-off, not a claim of universal dominance or a controlled timing speedup.

Size	Accuracy cost (pp)	Top-32 / LC GPU	GPU saved (%)	Observed time ratio	Params saved (%)	Top-32 gain recovered (%)
M	5.297	17.103	94.153	2.591	80.000	40.015
L	1.837	15.967	93.737	18.982	80.000	72.573

Unified dense construction overhead

All rows use the unified constructor; no dense-specialization construction times are substituted. Training includes evaluation.

Model	CPU construction (s)	Training (min)	GPU (GiB)	Construction / training (%)
MNIST	0.100	14.623	0.017	0.011
Fashion-MNIST	0.231	14.594	0.034	0.026
CIFAR-10 S	0.916	12.115	0.106	0.126
CIFAR-10 M	14.333	41.389	1.123	0.577
CIFAR-10 L	31.218	113.526	2.717	0.458

Dense first/deeper factorial: complete seed table

V, dense M at 20K updates. S retains LogicConnect; R preserves each predecessor degree in each input slot while randomizing pair identities. Native random is not the RR control.

Arm	Mean $\pm$ SD (%)	Seeds 0, 1, 2 (%)	Time (min)	GPU (GiB)
SS	59.133 $\pm$ 0.316	58.860, 59.480, 59.060	7.754	1.123
SR	58.613 $\pm$ 0.324	58.820, 58.780, 58.240	7.705	1.123
RS	54.640 $\pm$ 0.541	54.260, 55.260, 54.400	7.743	1.123
RR	54.927 $\pm$ 0.522	54.800, 54.480, 55.500	7.779	1.123

Dense factorial: all five contrasts

V, exploratory and unadjusted. First-layer effects have positive intervals. The additional deeper effect and interaction remain inconclusive.

Contrast	Gain (pp)	Paired 95% CI	Wins / 3	Raw gains (pp)
deeper given randomized first	-0.287	[-2.685, 2.111]	1	-0.540, 0.780, -1.100
deeper given structured first	0.520	[-0.523, 1.563]	3	0.040, 0.700, 0.820
first given randomized deeper	3.687	[1.621, 5.753]	3	4.020, 4.300, 2.740
first given structured deeper	4.493	[3.901, 5.086]	3	4.600, 4.220, 4.660
interaction	0.807	[-1.725, 3.338]	2	0.580, -0.080, 1.920

Dense M stage-selection controls

V, exploratory. The body/classifier effects condition on a fixed reference-ancestry classifier. WARP is an adapted recipe and trails the matched raw representation. These controls do not establish an isolated benefit from ancestry scoring.

Comparison	Gain (pp)	Paired 95% CI	n	Wins
LC minus random	4.313	[2.475, 6.152]	3	3
LC minus balanced random	0.487	[-0.795, 1.768]	3	3
LC minus nominal	-0.087	[-1.759, 1.585]	3	1

Convolutional S stage-selection controls

V, exploratory. The body/classifier effects condition on a fixed reference-ancestry classifier. WARP is an adapted recipe and trails the matched raw representation. These controls do not establish an isolated benefit from ancestry scoring.

Comparison	Gain (pp)	Paired 95% CI	n	Wins
LC minus random	1.707	[-2.600, 6.013]	3	3
LC minus balanced random	0.213	[-0.836, 1.263]	3	2
LC minus nominal	0.593	[-1.121, 2.308]	3	2

Convolutional body/classifier effects

V, exploratory. The body/classifier effects condition on a fixed reference-ancestry classifier. WARP is an adapted recipe and trails the matched raw representation. These controls do not establish an isolated benefit from ancestry scoring.

Comparison	Gain (pp)	Paired 95% CI	n	Wins
body with random head	1.207	[-3.169, 5.582]	3	3
body with LC head	2.140	[0.674, 3.606]	3	3
head with random body	-0.433	[-5.725, 4.858]	3	1
head with LC body	0.500	[-1.410, 2.410]	3	2
body head interaction	0.933	[-4.766, 6.633]	3	2

Dense M adapted-WARP effect

V, exploratory. The body/classifier effects condition on a fixed reference-ancestry classifier. WARP is an adapted recipe and trails the matched raw representation. These controls do not establish an isolated benefit from ancestry scoring.

Comparison	Gain (pp)	Paired 95% CI	n	Wins
LC minus random	4.700	[3.919, 5.481]	3	3

Convolutional S adapted-WARP effect

V, exploratory. The body/classifier effects condition on a fixed reference-ancestry classifier. WARP is an adapted recipe and trails the matched raw representation. These controls do not establish an isolated benefit from ancestry scoring.

Comparison	Gain (pp)	Paired 95% CI	n	Wins
LC minus random	0.087	[-1.243, 1.417]	3	1

Frozen CIFAR-10.1 transfer

T, no adaptation. One-time frozen evaluation. Single-seed convolutional points have no across-seed interval.

Model	LC gain (pp)	Paired 95% CI	n	Wins
dense_m	3.733	[2.801, 4.666]	3	3
conv_s	-0.050	–	1	0
conv_m	1.950	–	1	1

Third-round inventory, including negative reproductions

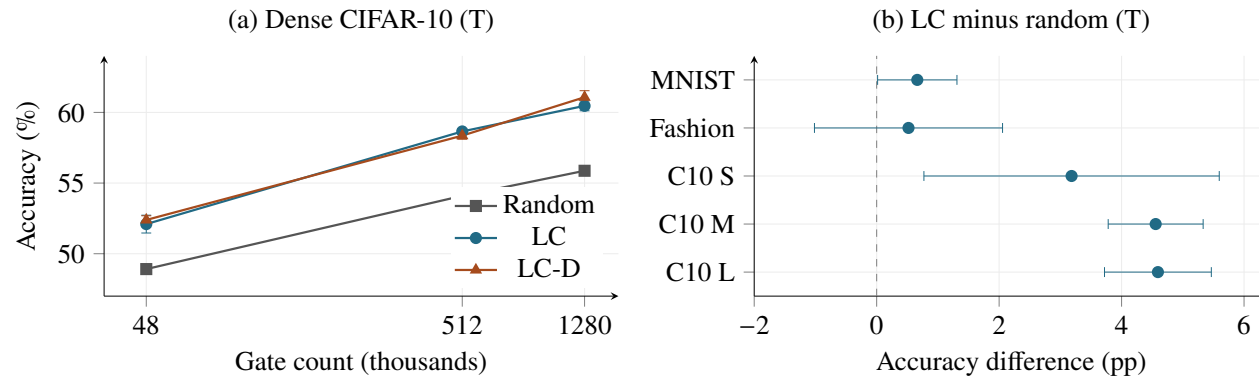
The BitLogic-related local reproductions failed to reproduce the published accuracy. They are preserved as negative evidence; low local comparator accuracy is not evidence of superiority over published BitLogic. Rank and protocol differences prevent topology-only attribution across arbitrary rows.

Protocol	Size	Method	Rank	n	Local T (%)	Time (min)	GPU (GiB)	Provenance
BitLogic-adapted	L	Learned rank-4	4	2	13.160 ± 0.240	164.470	9.456	reproduced-local
BitLogic-adapted	L	Random	2	2	25.160 ± 0.156	8.520	0.414	reproduced-local
BitLogic-adapted	L	LogicConnect	2	2	25.930 ± 2.871	8.561	0.414	our-transfer
BitLogic-adapted	M	Learned rank-4	4	2	16.625 ± 0.078	41.206	2.372	reproduced-local
BitLogic-adapted	M	Random	2	2	25.945 ± 0.870	8.582	0.104	reproduced-local
BitLogic-adapted	M	LogicConnect	2	2	26.040 ± 0.014	8.644	0.104	our-transfer
BitLogic-adapted	S	Learned rank-4	4	2	27.500 ± 0.113	9.921	0.611	reproduced-local
BitLogic-adapted	S	Random	2	2	26.175 ± 0.445	8.499	0.028	reproduced-local
BitLogic-adapted	S	LogicConnect	2	2	28.435 ± 1.648	8.472	0.028	our-transfer
LILogic	L	Random	2	3	55.333 ± 0.469	15.202	1.557	reproduced-local

Third-round inventory, including negative reproductions (continued)

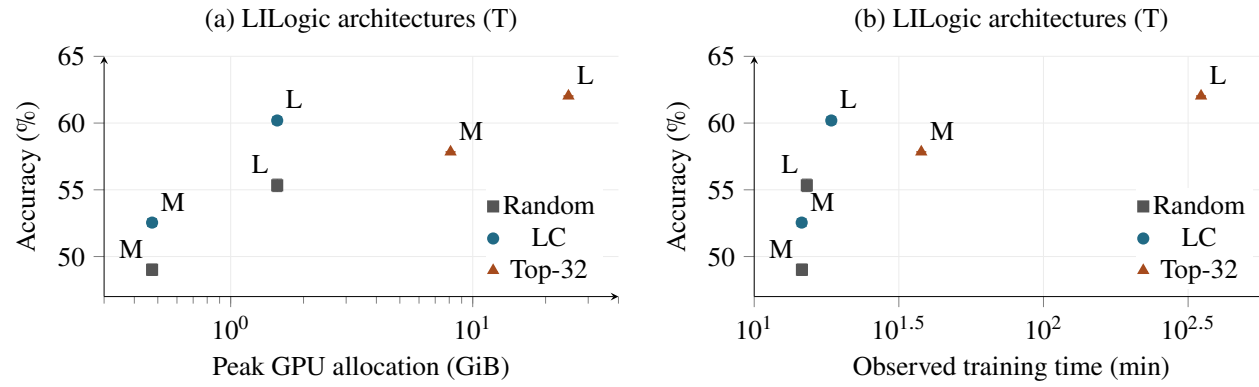
Protocol	Size	Method	Rank	n	Local T (%)	Time (min)	GPU (GiB)	Provenance
LILogic	L	Top-32	2	1	62.030	350.635	24.861	reproduced-local
LILogic	L	LogicConnect	2	3	60.193 $\pm$ 0.286	18.472	1.557	our-transfer
LILogic	M	Random	2	3	49.010 $\pm$ 0.426	14.622	0.474	reproduced-local
LILogic	M	Top-32	2	1	57.840	37.808	8.100	reproduced-local
LILogic	M	LogicConnect	2	3	52.543 $\pm$ 0.296	14.593	0.474	our-transfer
Original dense	L	LogicConnect	2	3	60.463 $\pm$ 0.348	113.526	2.717	our-transfer
Original dense	M	LogicConnect	2	3	58.653 $\pm$ 0.168	41.389	1.123	our-transfer

Dense Results



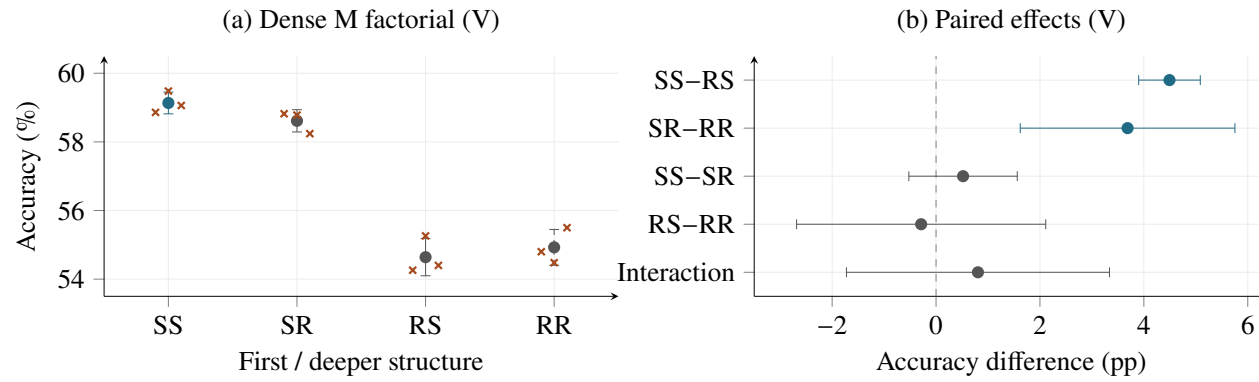
Dense accuracy at fixed budgets. (a) Mean $\pm$ sample SD over three seeds. LC-D is the separate dense specialization. (b) Paired 95% confidence intervals for LC minus random; all three CIFAR-10 intervals exclude zero.

Routing Tradeoff



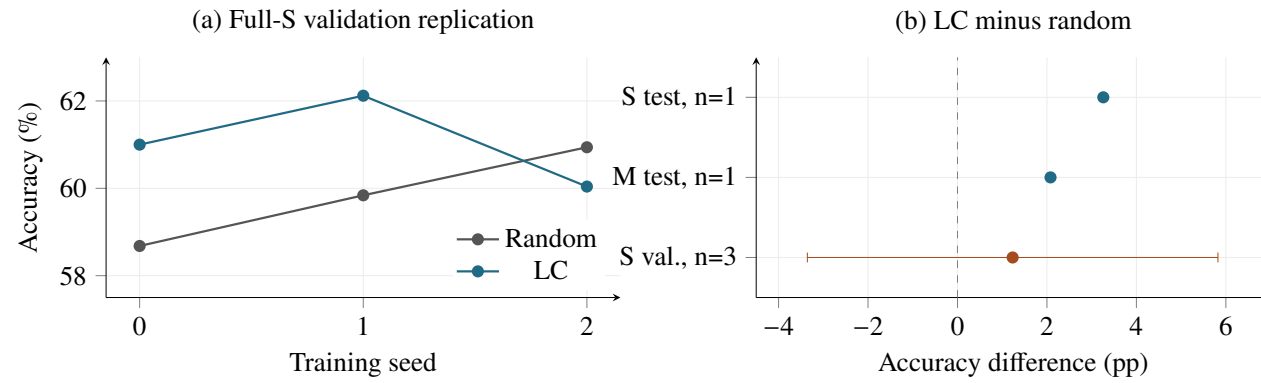
Accuracy versus (a) peak GPU allocation and (b) observed training time under the LILogic M/L configurations. LC and random: three seeds, mean $\pm$ sample SD. Top-32: one seed. M/L identify architectures; wall-time ratios are descriptive execution measurements.

Structural Ablation



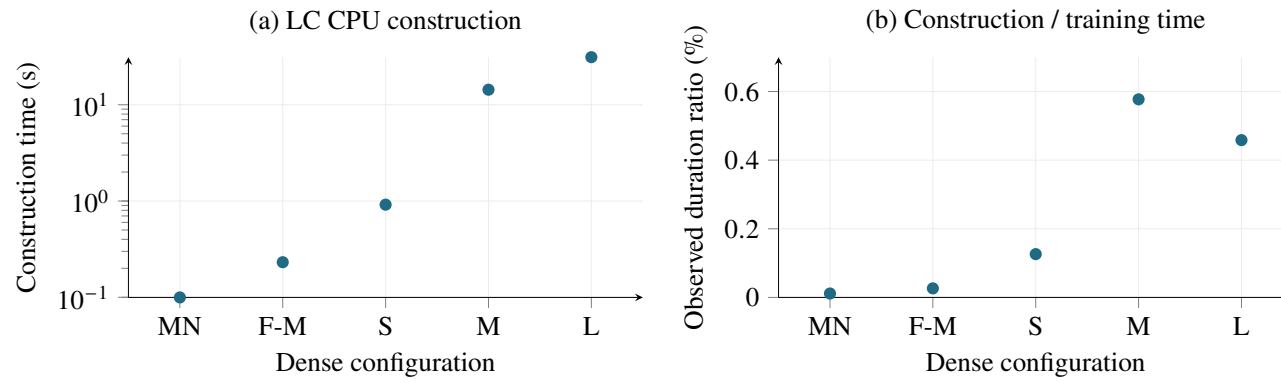
Dense M attribution at 20K updates. S retains LC structure; R randomizes pairs while preserving each predecessor degree in each input slot. (a) Three seed values (crosses) and mean $\pm$ sample SD. (b) All five paired 95% intervals, exploratory and unadjusted. The first two contrasts isolate first-layer structure.

Conv Replication



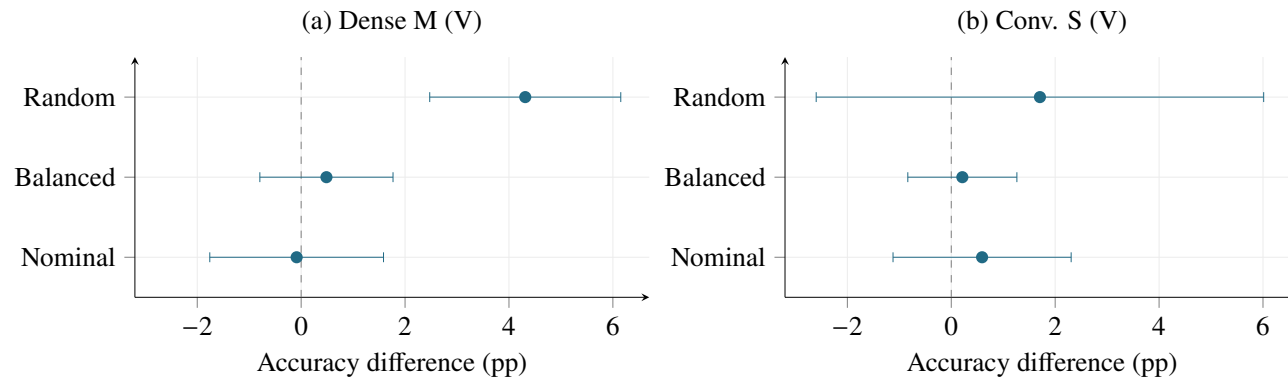
Convolutional replication and measurement scope. Single-seed tests have no error bars. Full-S validation uses three paired seeds with a 95% interval; its third seed reverses the sign of the gain. Validation and test observations are not pooled.

Construction Cost



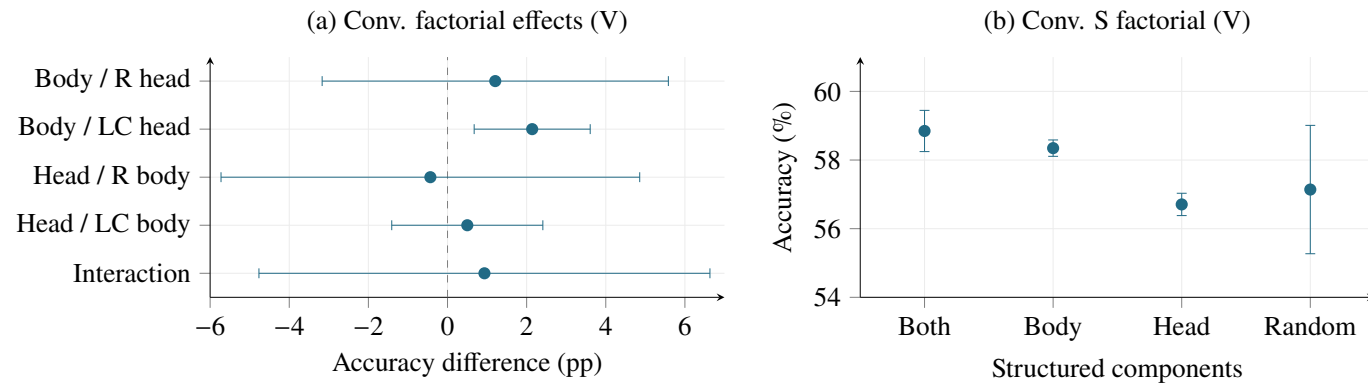
Offline LC construction for MNIST (MN), Fashion-MNIST (F-M), and CIFAR-10 S/M/L. Ratios use recorded training duration, including evaluation; they do not measure a controlled end-to-end speedup.

Mechanism Controls



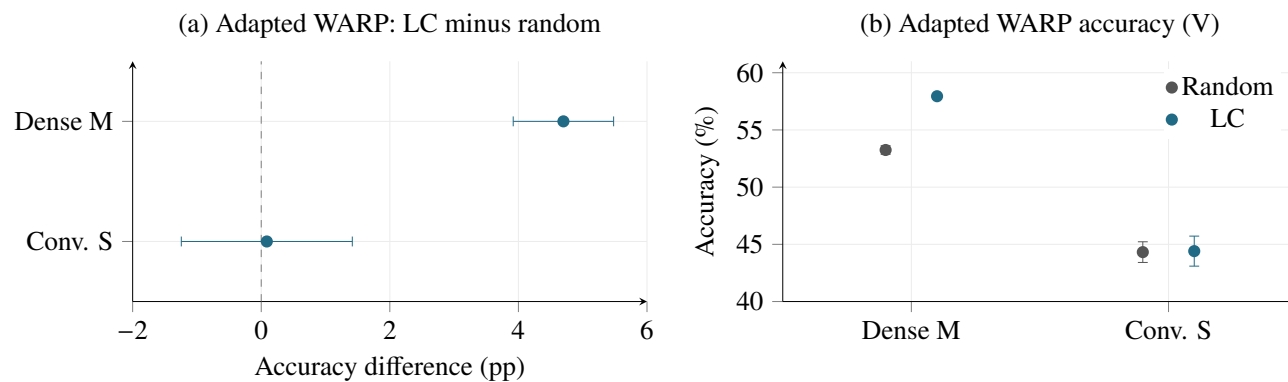
LC minus each stage-selection control at 20K updates, three paired seeds and exploratory 95% intervals. Neither coordinate establishes an incremental benefit over the nominal schedule. The balanced control is distinct from native random wiring.

Conv Factorial



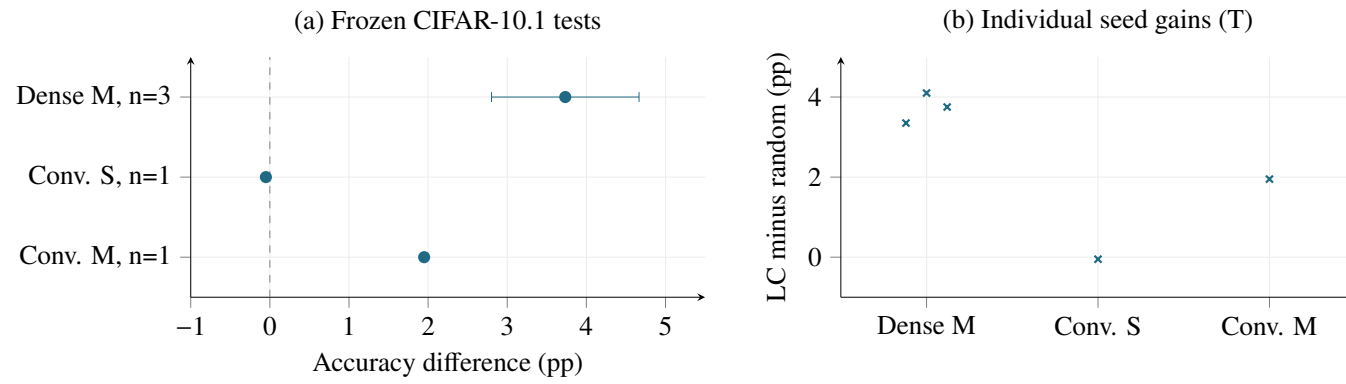
Convolutional body/classifier attribution at 20K updates, three seeds. The structured classifier retains reference-body ancestry, so body effects are conditional on that classifier. (a) All five exploratory paired 95% intervals. (b) Mean $\pm$ sample SD.

Warp Compatibility



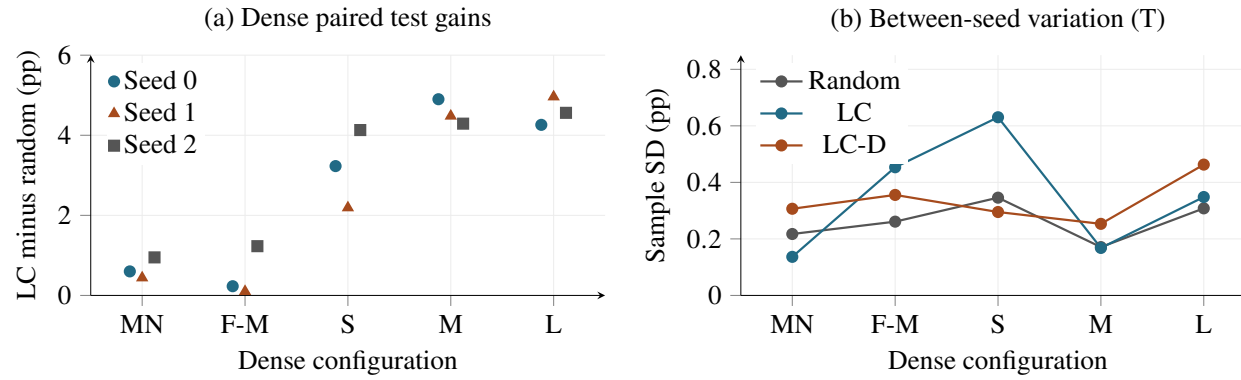
Compatibility with adapted WARP training. Dense M uses 108K updates; convolutional S uses 20K. Three seeds; paired 95% intervals in (a), mean $\pm$ sample SD in (b). Both adapted recipes trail matched raw training; these data do not reproduce WARP's published headline results.

Transfer



Frozen CIFAR-10.1 evaluation without adaptation or repeat inference. Dense M has a three-seed paired 95% interval; convolutional points each use one seed. (b) All available paired seed differences.

Dense Seed Gains



All 15 paired dense LC test differences and the sample standard deviations of all three fixed constructions. MN/F-M denote MNIST/Fashion-MNIST; S/M/L denote CIFAR-10 sizes. Positive seed differences do not guarantee that every three-seed confidence interval excludes zero.